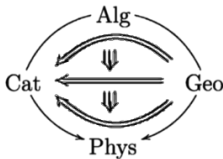


Algebraic Topology: Beyond Singular Cohomology

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¹webpage - nLab page

Plain

1. General Ideas

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“What is Algebraic Topology?”

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2. Abstract Invariants

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General Ideas

What is Algebraic Topology?



- ▶ Algebraic Topology is about classification of spaces up to homotopy;



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Algebraic Topology is about homotopy invariants.



General Ideas

What is Algebraic Topology?

1. Classification Problem and Invariants



General Ideas

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1. Classification Problem and Invariants
2. Functorial Invariants



General Ideas

What is Algebraic Topology?

1. Classification Problem and Invariants
2. Functorial Invariants
3. Weak Equivalences



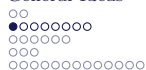
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What is Algebraic Topology?

1. Classification Problem and Invariants
2. Functorial Invariants
3. Weak Equivalences
4. Homotopy Invariants

1. Classification Problem and Invariants

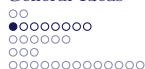
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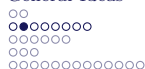
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 - ▶ how objects *interact* with each the other.
- ▶ When studying an area o math we think of it as something “new”. The basic problem is then the *classification problem*:

Classification Problem: Can we describe the objects of an area o math in terms of already known structures?

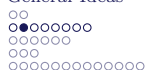
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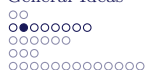


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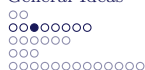
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- ▶ typically, an area of math comes equipped with a notion of *equivalence* or *isomorphism*, denoted by $\simeq$
- ▶ in this case one typically looks at the classification problem *up to equivalences/isomorphisms*, meaning searching for bijections

$$F : \text{Ob} / \simeq \rightarrow A$$

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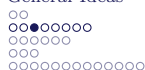
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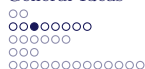
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- ▶ But in order to get such bijections one first needs to find a function F and then prove that they are indeed bijective...

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Definition. An A -invariant in an area of math is a map $f : \text{Ob} \rightarrow A$ which pass to the quotient, as below.

$$\begin{array}{ccc}
 \text{Ob} & \xrightarrow{f} & A \\
 \downarrow & & \downarrow \\
 \text{Ob} / \simeq & \xrightarrow{f} & A / \simeq
 \end{array}$$



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Conclusion. Every invariant of an area of math classifies a “piece” of that area.

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- ▶ indeed, for all of them $\mathbb{S}^2$ and $\mathbb{S}^1 \times \mathbb{S}^1$ belong to the same component, so that one cannot conclude that they are non-homeomorphic...

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 - ▶ *limits*: taking arbitrary products between objects, kernels of morphisms or, more generally, taking the incidence space of two morphisms (pullbacks), projective limits, etc.



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- ▶ varying the shapes J we get different limits/colimits. When exist, they are denoted by $\lim(J)$ and $\operatorname{colim}(J)$.

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typically categories have products and coproducts. Thus, they fail to be complete/cocomplete because they does not have certain equalizazers/coequalizers.

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- ▶ A class $[f] : X \rightarrow Y$ in $\mathbf{C}/\sim$ is an isomorphism iff it is represented by some morphism $f : X \rightarrow Y$ in $\mathbf{C}$ for which there exists $g : Y \rightarrow X$ such that

$$g \circ f \sim_{XX} id_X \quad \text{and} \quad f \circ g \sim_{YY} id_Y.$$

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4. Homotopy Invariants

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 - ▶ invariants of other areas of math are, secretly, invariants from Algebraic Topology
 - ▶ there should exist an area of math which studies these “abstract homotopy invariants”. This is *Abstract Homotopy Theory*.

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Notation. From now on

$$[X; -] : \mathbf{C}/\sim \rightarrow \mathbf{Set} \quad \text{and} \quad [-; X] : \mathbf{C}^{op}/\sim \rightarrow \mathbf{Set}$$

will denote the hom-functors of $\mathbf{C}/\sim$.

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- ▶ if limits are about commutative digrams, homotopy limits are about diagrams which commute only up to $\sim$
- ▶ homotopy limits of shapes $J : \mathbf{J} \rightarrow \mathbf{C}$ can be computed as usual limits, but first we need to replace J by some more well-behaved versions LJ and RJ , called *left/right-derived functors* of J .



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Conclusion. The canonical homotopy invariants $[-; X]$ and $[X; -]$ are computable.

Abstract Invariants

What is Cohomology?

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- ▶ the main example is **Top** with the homotopy category of topological spaces.

- Motivated by the previous discussion, the strategy to attack the classification problem of $\mathrm{Ho}(\mathbf{C})$ is to look at the homotopy invariants defined by hom-functors $[X; -]$ and $[-; X]$.

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- ▶ A monoid (resp. comonoid, group, etc.) in a homotopy category $\mathrm{Ho}(\mathbf{C})$ is usually called H -monoid, (resp. H -comonoid, H -group).

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- ▶ Thus, for every X , $\pi_0(X; \mathbb{S}^n) = [\mathbb{S}^n; X]$ is a group if $n > 0$ and an abelian group if $n \geq 2$. we write $\pi_n(X)$ instead $\pi_0(X; \mathbb{S}^n)$ and say that this is the n -th *homotopy group* of X .
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- ▶ an ∞ -loop space object is an object X which has n -deloopings for every n .

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- **Example.** In **Top**, if G is an abelian group, then B^n exists for every n . The cohomology theory defined by corresponding ∞ -loop space is precisely singular cohomology with coefficients in G .

Thanks